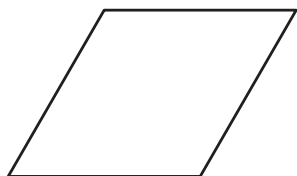
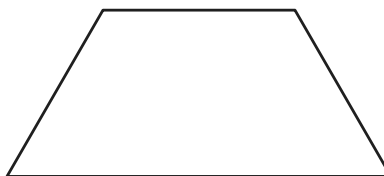


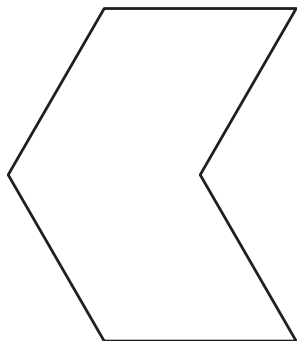
Problem 1: Cover each shape below with green triangles only. No gaps, no overlaps, nothing sticking out. Count the greens on the shape. Say the number, and your grown-up writes it in the box next to the shape.



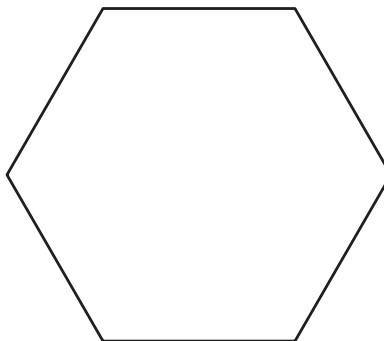
greens



greens

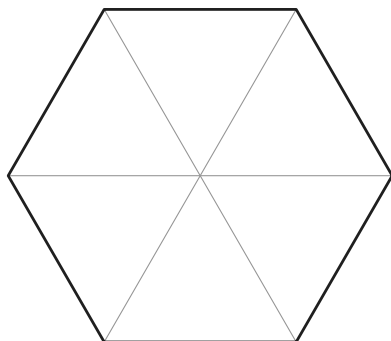


greens

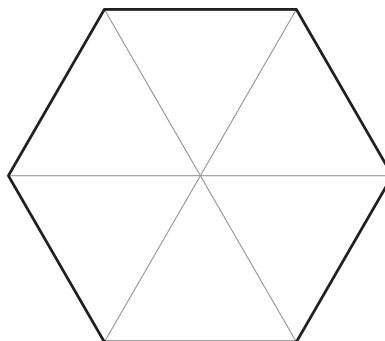


greens

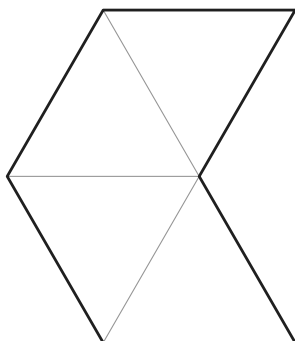
Problem 2: Cover the hexagon (the shape of a yellow piece) with blue pieces only. How many blues? Cover it again with red pieces only. How many reds? Then cover the chevron (the shape of a purple piece) with blue pieces only. How many blues?



blues

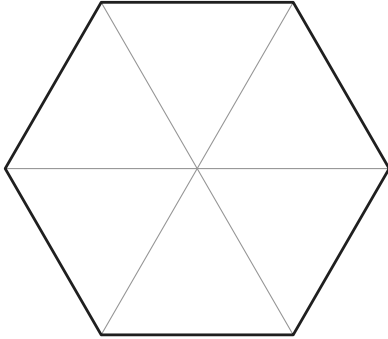


reds

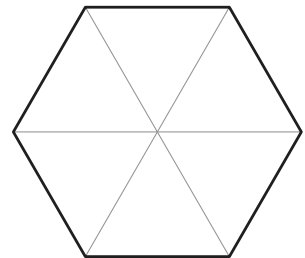
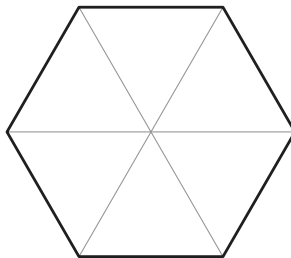
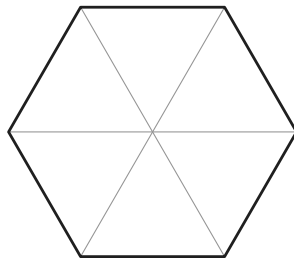
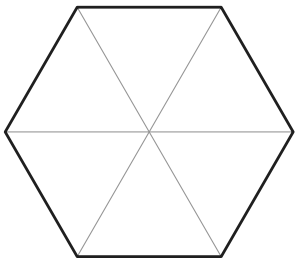


blues

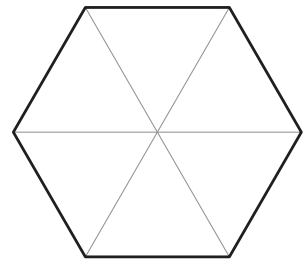
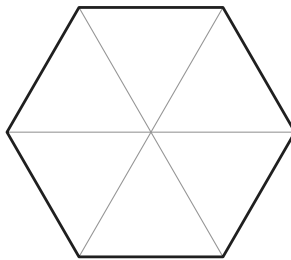
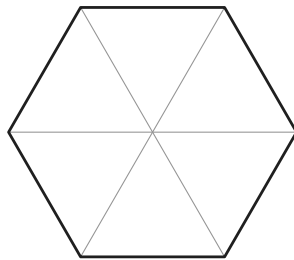
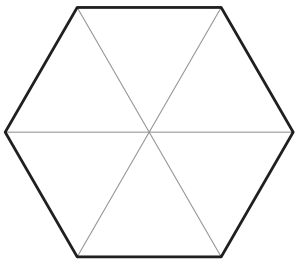
Problem 3: Cover the big hexagon with blue pieces only. Then take the blues off and cover it a different way, with blues only. Two ways are different if a blue lies in a different place on the hexagon. Show each way on a small hexagon: colour each piece with a different crayon. Say how many ways you found.



ways with blues

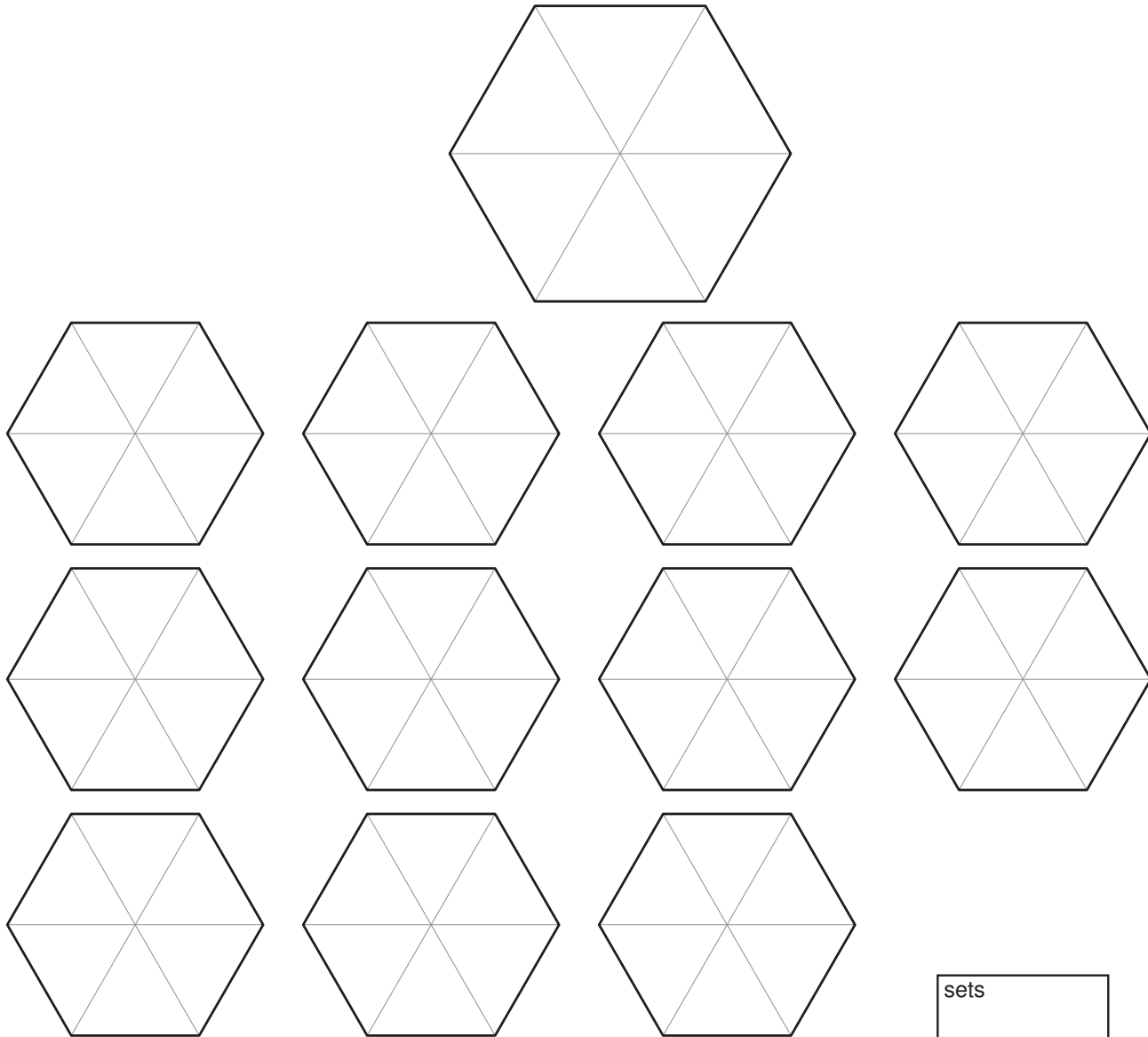


Problem 4: Cover the big hexagon with red pieces only. Find every way. Show each way on a small hexagon, one crayon colour for each piece. Say how many ways.



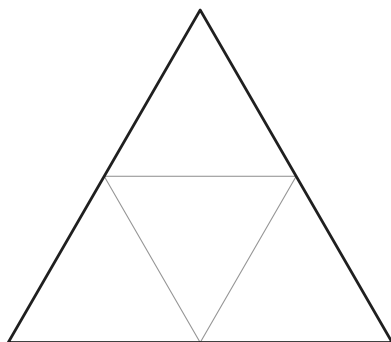
ways with reds

Problem 5: Fill the big hexagon with any pieces you like: green, blue, red, yellow or purple. Look at which pieces you used. Colour one small hexagon to show them: green where a green lies, blue where a blue lies, red for red, yellow for yellow, purple for purple. Now fill the big hexagon with a different set of pieces and colour another small hexagon. Two fillings that use the same pieces are the same set, even if the pieces lie in different places: colour that set only once. Find every set you can. Say how many sets.



sets

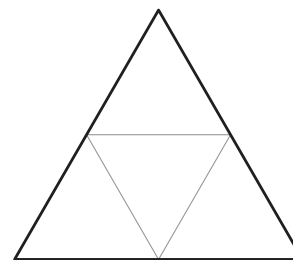
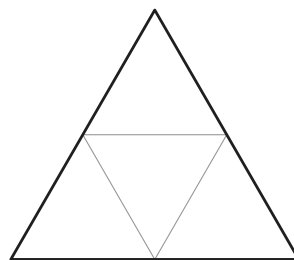
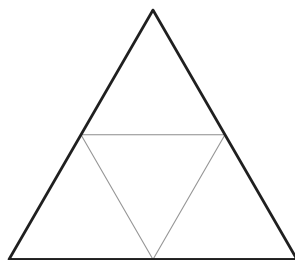
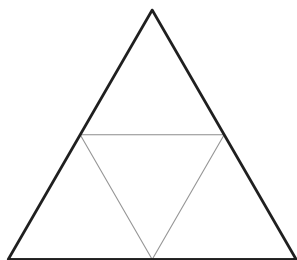
Problem 6: Cover the big triangle with greens only. How many greens? Now take them off and try to cover it with blues only. Can you do it? Say yes or no.



greens

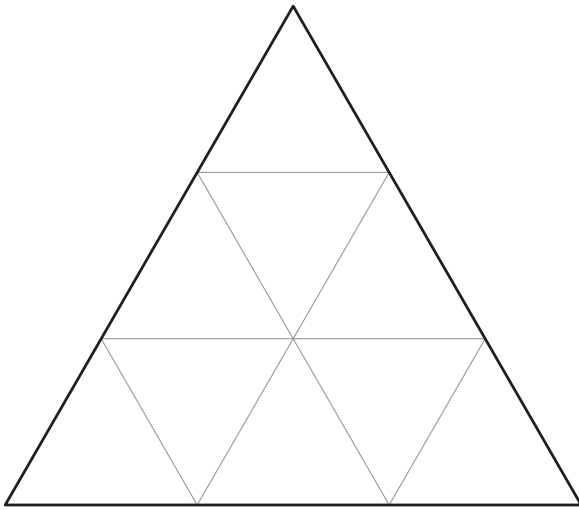
blues only **YES** **NO**

Problem 7: Cover the big triangle with one red piece and some greens. Show it on a small triangle: colour red where the red lies and green where the greens lie. Now move the red to a different place and cover the triangle again. Find every way. Say how many ways.



ways

Problem 8: Cover the big triangle with greens only. How many greens? Take them off and try blues only. Can you? Say yes or no. Take them off and try reds only. Can you? Say yes or no.

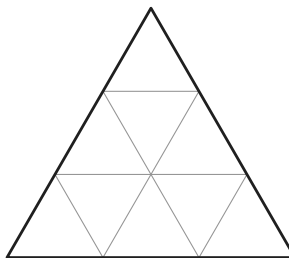
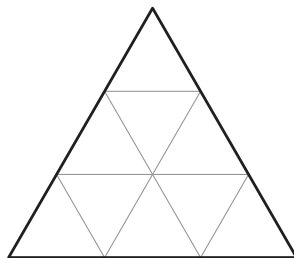
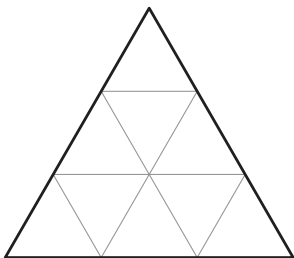


greens

blues only **YES** **NO**

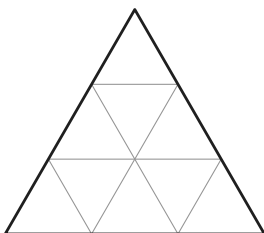
reds only **YES** **NO**

Problem 9: Cover the big triangle with reds only. Show it on a small triangle, one crayon colour for each red. Take the reds off and find a different way with reds only. Say how many ways there are.



ways

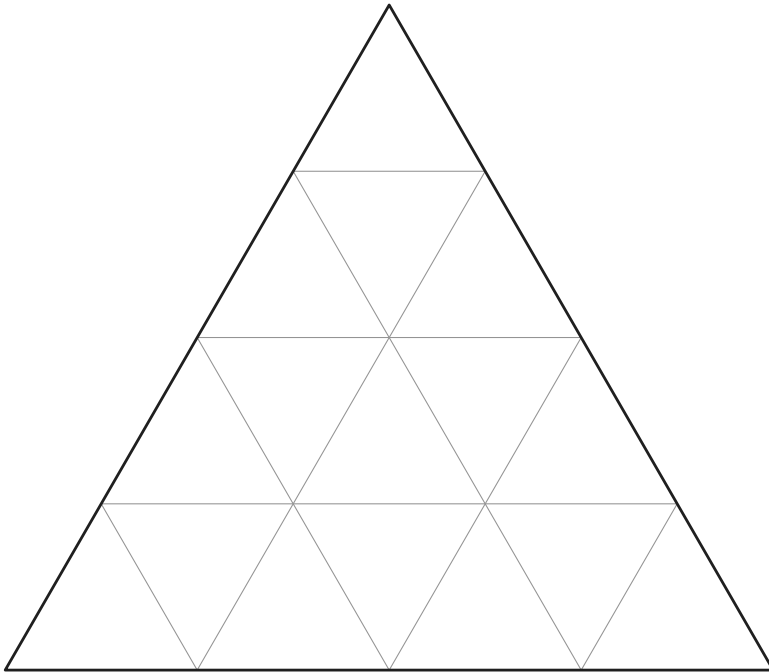
Problem 10: Put one yellow on the big triangle of Problem 8 so that it fits inside the outline. Where can it go? Now try to put two yellows on at once. Can you? Say yes or no. Leave one yellow on and fill the rest with greens. How many greens? Colour the small triangle to show it.



two yellows **YES** **NO**

greens

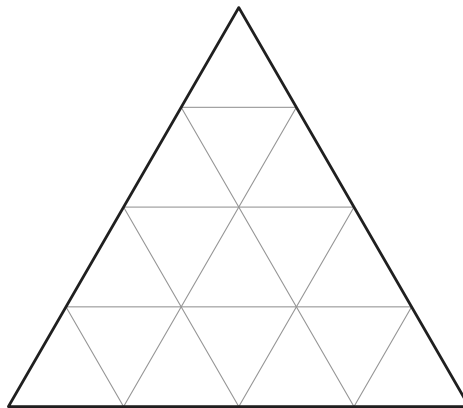
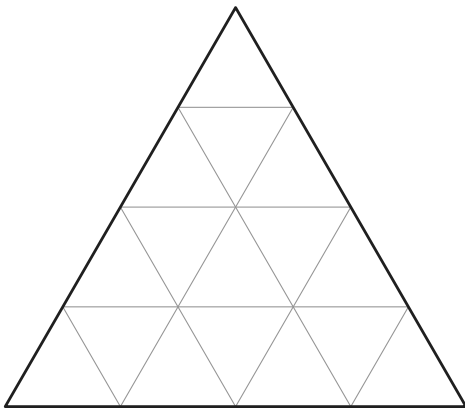
Problem 11: Try to cover the big triangle with blues only. Can you? Say yes or no. Take them off and try reds only. Can you? Say yes or no.



blues only **YES** **NO**

reds only **YES** **NO**

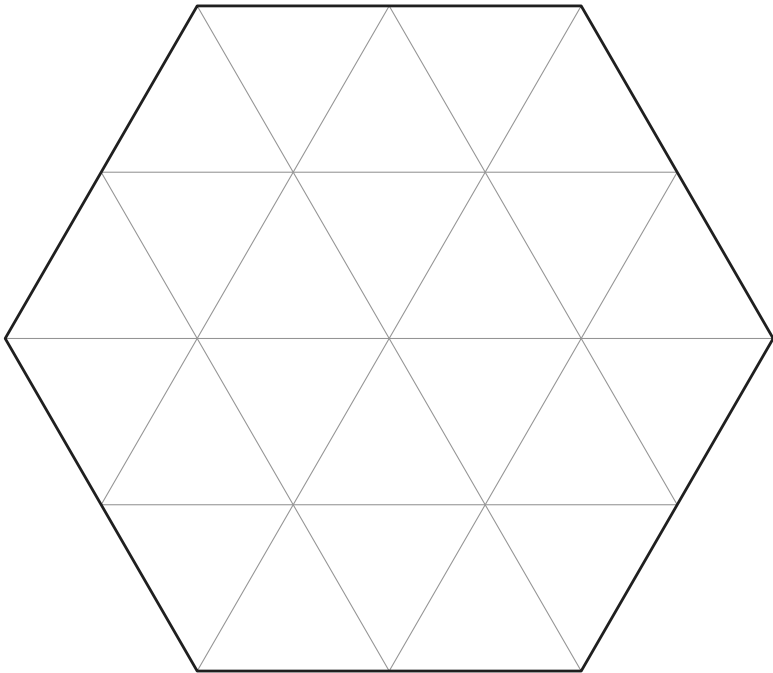
Problem 12: Cover the big triangle of Problem 11 without using any green piece. Look at which pieces you used. Colour the first small triangle: red where a red lies, blue for blue, purple for purple, yellow for yellow. Count your reds. Then take everything off and find a way that uses a different set of pieces, still with no green. Colour the second small triangle and count the reds again.



reds, 1st way

reds, 2nd way

Problem 13: Cover the big hexagon with blue pieces only. How many blues? Now look at each blue. It is standing, or leaning left, or leaning right, like the three pictures. Count the standing blues, the leaning-left blues and the leaning-right blues, and your grown-up writes the three numbers in the first row of the table. Take the blues off, cover the hexagon a different way, and count again in the second row. Do it a third time.



standing



leaning left



leaning right

blues in all

	standing	leaning left	leaning right
first way			
second way			
third way			

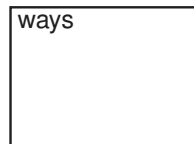
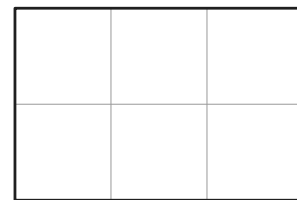
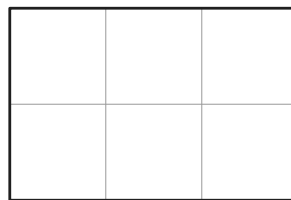
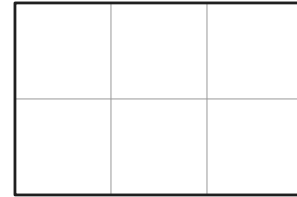
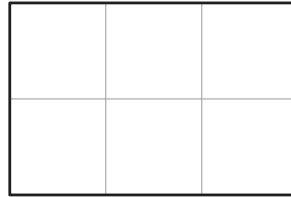
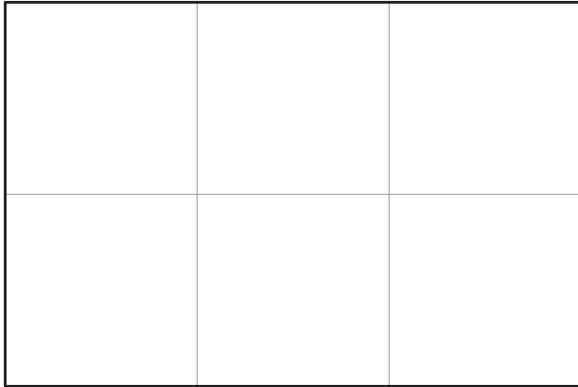
Problem 14: Cover the big hexagon with red pieces only. How many reds? Then try yellow pieces only. Put down as many yellows as will fit without overlapping. How many did you fit? Can you cover the whole hexagon with yellows? Say yes or no.

reds

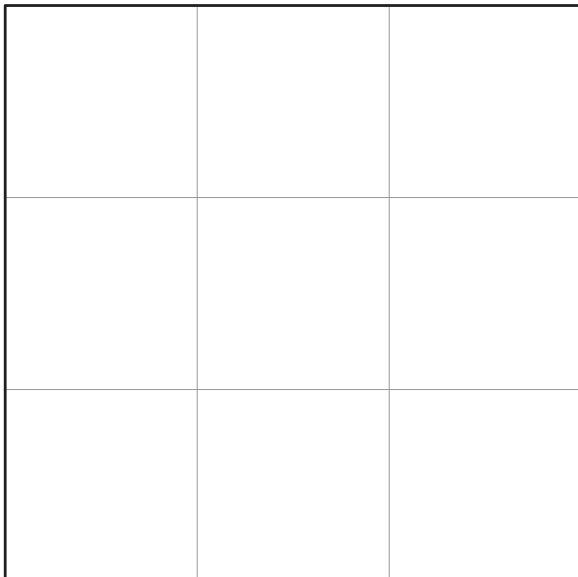
yellows that fit

all yellow **YES** **NO**

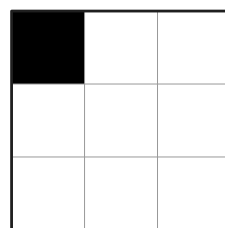
Problem 15: A paper domino covers two squares that share a side. Cover the big board with dominoes. No gaps, no overlaps, nothing sticking out. Find every way. Show each way on a small board: draw one thick line through the two squares of each domino. Say how many ways.



Problem 16: Try to cover the big square board with dominoes. Can you? Say yes or no. Now put the black paper square on the big board where the first small picture shows it. A domino may never lie on the black square. Cover the eight white squares with dominoes. Can you? Say yes or no. Do the same for the second picture and the third picture.

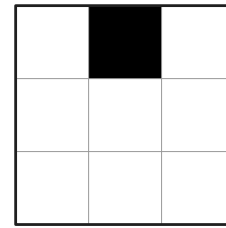


whole board **YES NO**



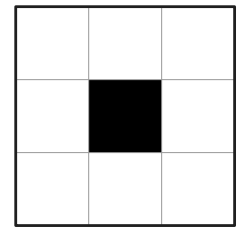
corner

YES NO



side

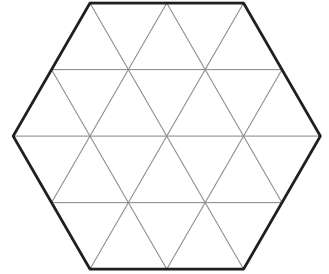
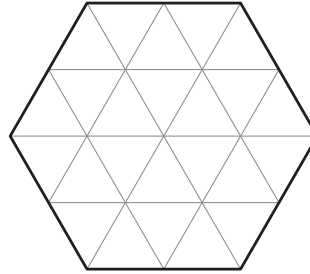
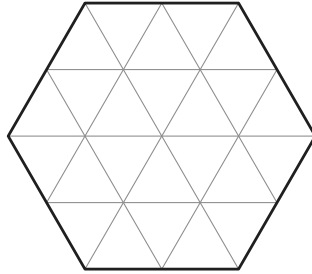
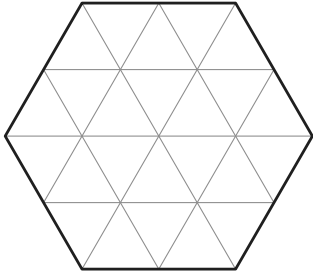
YES NO



centre

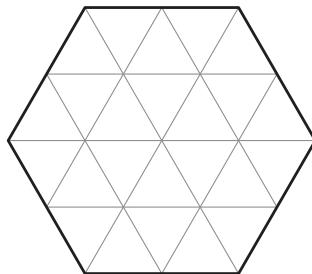
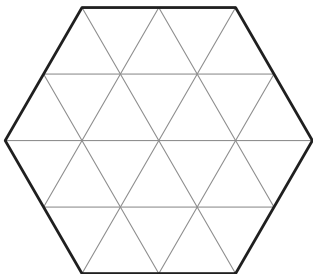
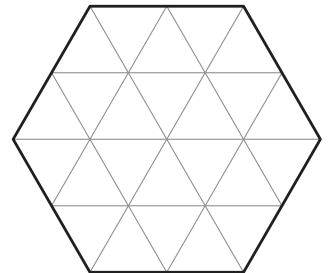
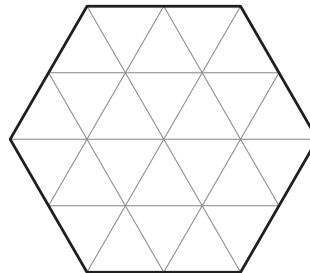
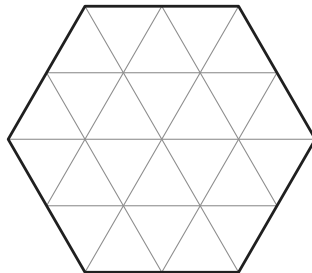
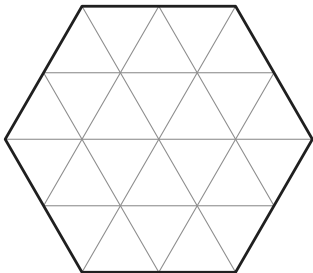
YES NO

Problem 17: Take two greens. Put them together so that a whole side touches a whole side. What shape do you get? Now take three greens and make a shape the same way. Then take four greens. Make a shape, show it to your grown-up, and your grown-up draws it on one of the hexagon grids below. Now make a shape with four greens that is different: not the same shape turned around or flipped over. Find every different shape four greens can make. Say how many.



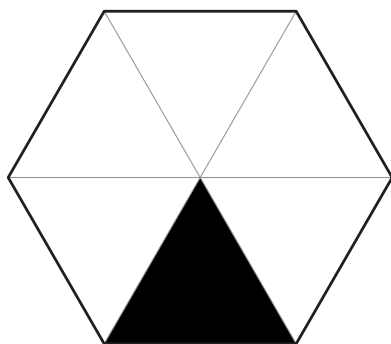
shapes from 4 greens

Problem 18: Now take five greens. Make a shape with all five, whole side to whole side, and your grown-up draws it. Find every different shape five greens can make. Say how many.



shapes from 5 greens

Problem 19: This hexagon has one triangle bitten off; the black part is not there. Cover the white part with any pieces you like. Colour a small copy to show which pieces you used, with the colour of each piece. Find every different set of pieces, and colour each set once. Then try blues only: can you? Say yes or no. Reds only: can you? Say yes or no.

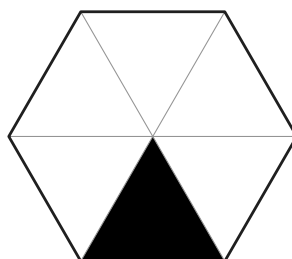
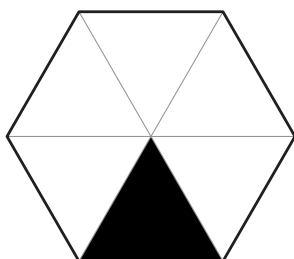
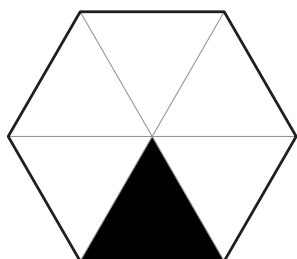
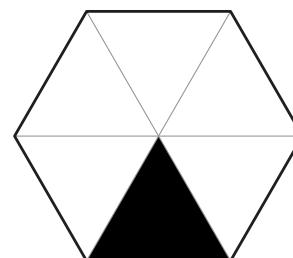
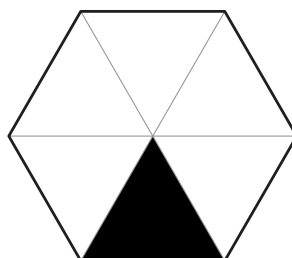
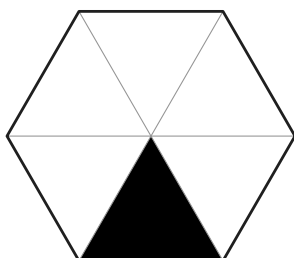
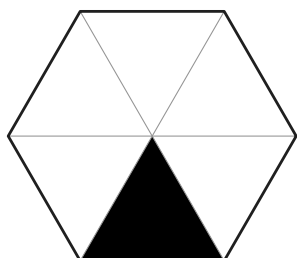


blues only **YES**

NO

reds only **YES**

NO



sets